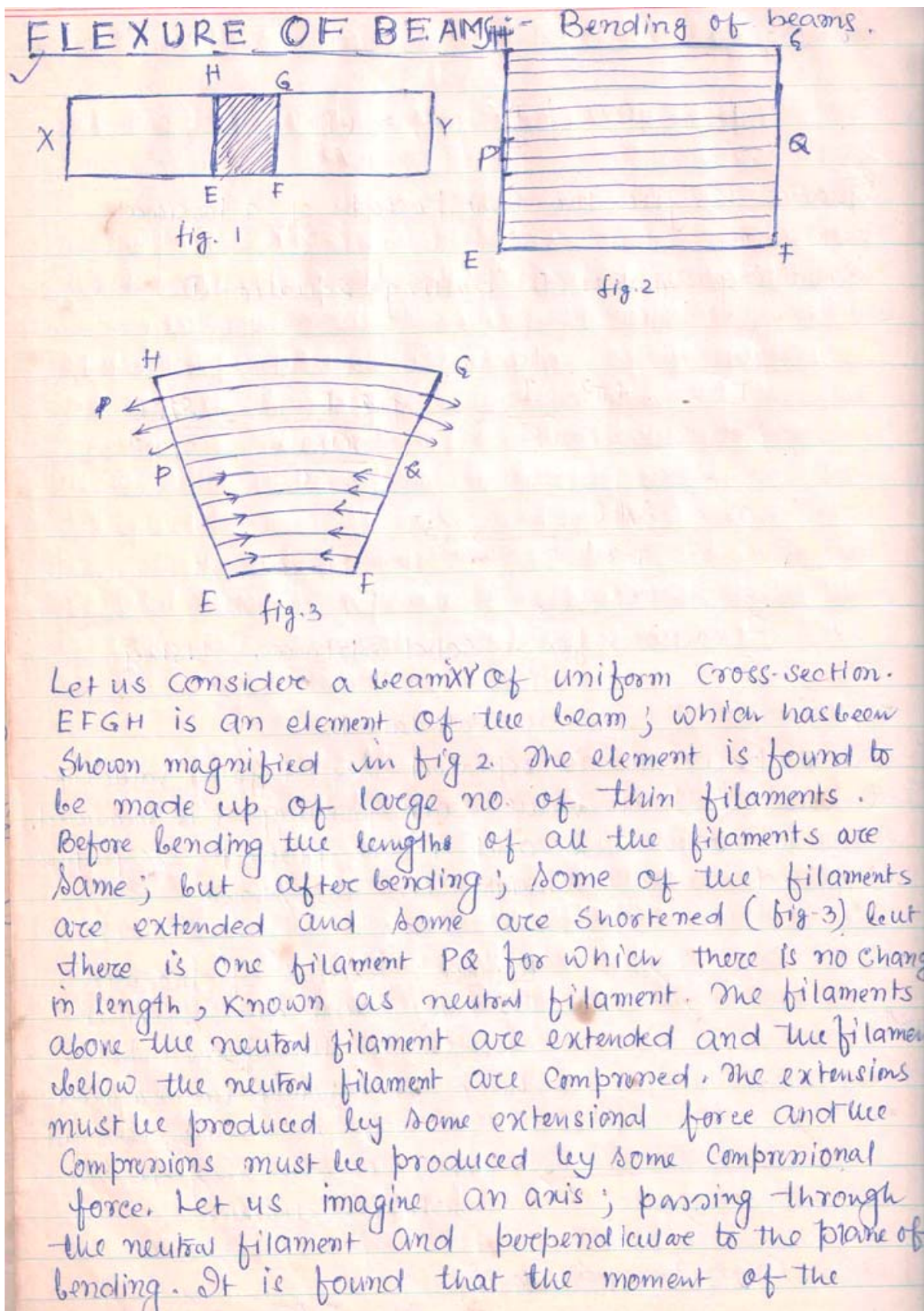


Flexure Of Beam



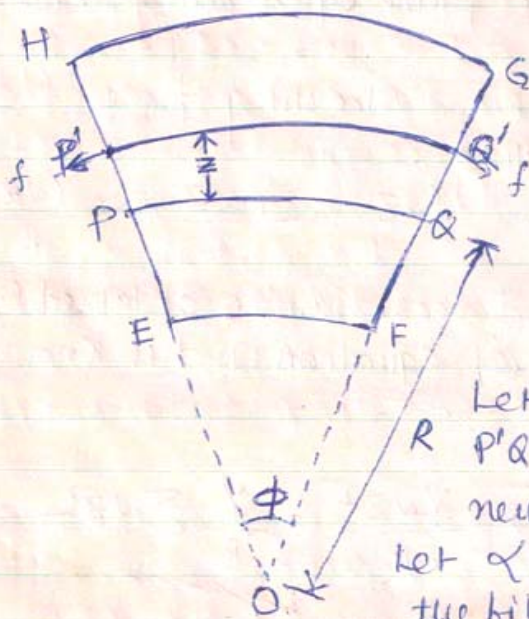


Flexure Of Beam

extensional, as well as compressional forces about the axis of bending are same in direction.

The sum total of moments of all the forces, extensional as well as compressional about the axis of bending is known as Bending moment.

We now find the Bending Moment:



Let 'O' be the Centre of Curvature of the bent element.

Δl = length of the filament PQ

R = Radius of curvature of the neutral filament PQ

Let us consider a filament $P'Q'$ at a distance z above the neutral filament.

Let α = area of cross section of the filament $P'Q'$

Let f = Stretching force, acting on the filament

Before bending length of filament $P'Q' = \Delta l$ = the length of the neutral filament.

from PQ : $\phi = \frac{\Delta l}{R}$ or $\boxed{\Delta l = R\phi}$

Let $\Delta l'$ = length of the filament $P'Q'$ after bending.
from $P'Q'$: $\phi = \frac{\Delta l'}{(R+z)}$ or $\Delta l' = (R+z)\phi$

$\therefore$ Change in length = $\Delta l' - \Delta l = (R+z)\phi - R\phi$
 $= z\phi$



Flexure Of Beam

Extensional strain: $\frac{Z\phi}{R\phi} = \frac{Z}{R}$

The stress = $\frac{f}{\alpha}$ $\therefore$ Young's modulus = $Y = \frac{f/\alpha}{Z/R}$

$$\text{or } f = \frac{Y\alpha Z}{R}$$

Imagine an axis passing through the point P and perpendicular to the plane of bending, known as axis of bending.

Taking the moment of the stretching force 'f' about the axis of bending;

We get the moment = $f \cdot Z = \frac{Y\alpha Z}{R} \cdot Z = \frac{Y\alpha Z^2}{R}$

The total moment of all forces can be obtained by taking the summation of equation (1) and is known as Bending Moment. M.

$$\text{Bending moment } M = \sum \frac{Y\alpha Z^2}{R} = \frac{Y}{R} \sum \alpha Z^2 \rightarrow (2)$$

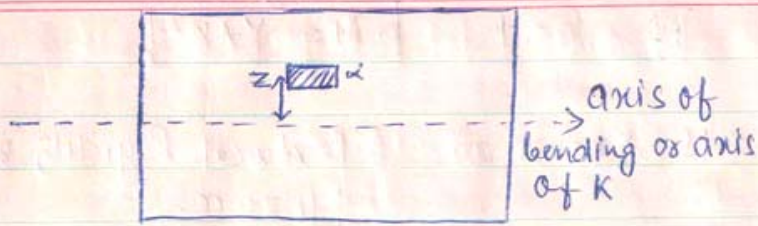
Put $\sum \alpha Z^2 = Ak^2 \rightarrow (3)$ Known as geometrical M.I. of the beam; where A is the area of cross-section of the beam and K is the radius of gyration about the axis of bending.

$$\text{B.M} = \frac{YAk^2}{R} \rightarrow (4)$$

Discussions : Meaning of the term Geometrical M.I. :-



Flexure Of Beam



The rectangle represents the section of the beam (which is perpendicular to the plane of the paper) by the plane of the paper.

The beam is bent in a vertical plane, perpendicular to the plane of the paper.

The dotted line represents an axis passing through the centre of the cross section and perpendicular to the plane of bending and hence is the axis of bending also known as axis of K.

Given: A = Area of the rectangle.
 σ = Mass per unit ^{area} length of the rectangle.

Consider an element of area ' d ' at a distance ' z ' from the axis of K

M.I of the element about the given axis $= (\sigma d) \cdot z^2$

$\therefore$ M.I of the whole lamina about the axis

$$= \sum \sigma d z^2$$

$$= \sigma \sum d z^2 \rightarrow (a)$$

But M.I. of the lamina about the given axis $= M K^2$

$$= \sigma A K^2 \rightarrow (b)$$

Equating (a) and (b) :-

$$\cancel{\sigma} \sum d z^2 = \cancel{\sigma} A K^2$$

$$\boxed{\sum d z^2 = A K^2} = \text{Geometrical M.I.}$$



Flexure Of Beam

②. $M = \frac{YAK^2}{R}$ if $R=1$, $M = YAK^2$

The term YAK^2 is known as Flextural Rigidity or flexure.

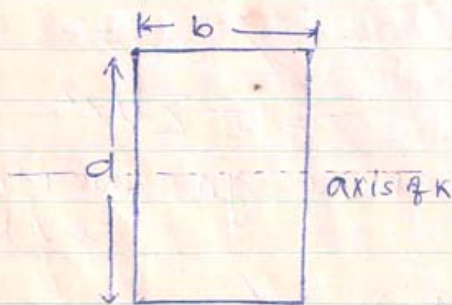
Hence the flexure of a beam can be defined as the Bending moment required to bend the beam, with unit radius of curvature. The flexure of a beam is a measure of the strength of the beam. Greater is the value of Flextural rigidity; greater will be the moment required to bend the beam and hence greater will be the strength of the beam.

The flextural rigidity i.e. the strength of the beam, depends on the geometrical M.I of the beam.

We now calculate the geometrical M.I for beams of various cross-section:-

(i) beam of rectangular cross-section:-

Given: b and d = the length of the horizontal and vertical side of the cross-section respectively.



Area of the plane $A = b \cdot d$
M.I of the rectangle about the given axis

$$I = Mk^2 = \frac{Md^2}{12}$$

$$\therefore k^2 = \frac{d^2}{12}$$

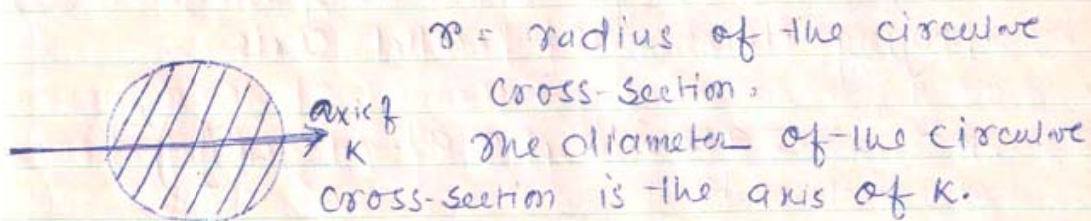
$$\text{Geometrical M.I} = AK^2 = b \cdot d \cdot \frac{d^2}{12} = \frac{bd^3}{12}$$



Flexure Of Beam

Since 'd' occurs in the power 3, hence AK^2 will be very large if we increase d. Hence the beams are placed with the length of the vertical side of cross-section higher.

(ii) Beam of circular cross-section:



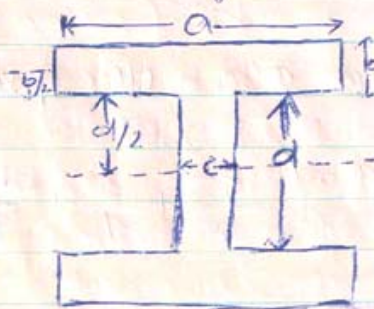
M.I of the circular disc about its diameter

$$I = \frac{Mr^2}{4} = Mk^2$$

$$\therefore k^2 = \frac{r^2}{4}$$

$$\text{Geometrical M.I} = AK^2 = \pi r^2 \cdot \frac{r^2}{4} = \frac{\pi r^4}{4}$$

(iii) Beam of I - cross section:-



let a and c = the length of the horizontal side of the section.

b & d = the length of the vertical sides of the section.

The axis of bending is shown by the dotted lines

Let us consider the rectangle of area $a \times b$.
M.I of this rectangle; about the axis passing through the centre and $\perp$ to the



Flexure Of Beam

the side 'b' i.e. parallel to the given axis of K

$$= \frac{\text{mass} \cdot b^2}{12}, \quad \frac{\sigma ab \cdot b^2}{12}$$

σ = mass per unit area.

$\therefore$ M.I of the rectangle about the given axis of K;
by applying theorem of parallel axis

$$\sigma ab \cdot K^2 = \frac{\sigma ab \cdot b^2}{12} + \sigma ab \left(\frac{b}{2} + \frac{d}{2} \right)^2$$

$$\text{G.M.I. } A K^2 = ab \left[\frac{b^2}{12} + \frac{b^2}{4} + \frac{d^2}{4} + \frac{bd}{2} \right]$$

M.I of the rectangle of area (cd) about the given axis of K:

$$\sigma cd \cdot K^2 = \sigma cd \cdot \frac{d^2}{12}$$

$$\text{G.M.I. } (A K^2) = \frac{cd \cdot d^2}{12}$$

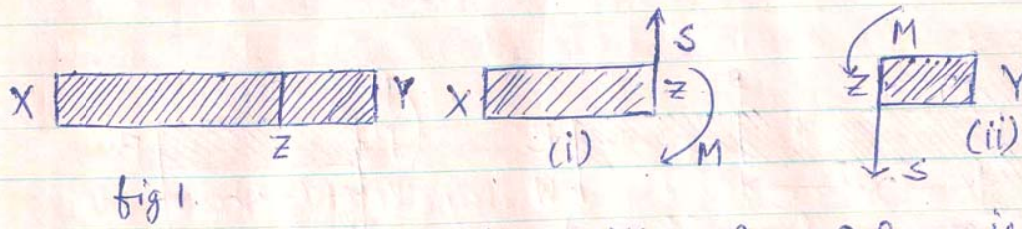
Total M.I of the I-Section about the given axis:-

$$\text{G.M.I.} = \left[\frac{cd^3}{12} + 2ab \left(\frac{b^2}{12} + \frac{b^2}{4} + \frac{d^2}{4} + \frac{bd}{2} \right) \right]$$

Flexure Of Beam



To solve Problems of in Bending of beam:-



Let us consider a beam XY. When a beam is bent a system of forces act on the beam which is equivalent to (i) a force & (ii) a couple. The force is known as Shearing force S and the couple is known as Bending moment M .

To find the Shearing force and B.M at any point say Z in the beam, we imagine a section of the beam through that point and the two portions of the beam are then shown separated. The convention we follow is this:

The right hand portion (ii) exerts on the left hand portion (i) a shearing force along upward direction and a bending moment M along clockwise direction.

The left hand portion (i) exerts on the right hand portion (ii) a shearing force S along downward direction and a bending moment M along anti-clockwise direction.